THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE UNIT TEST SET B

Chapters 10–13: **Light: Mirrors and Lenses, Keeping Time with the Skies, How Nature Works in Harmony, Our Home Earth, a Unique Life-Sustaining Planet** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. A magnifying glass is a: 1

(a) convex lens **(b)** concave lens **(c)** convex mirror **(d)** plane mirror

2. If a ray falls along the normal, the angle of reflection is: 1

(a) 180° **(b)** 0° **(c)** 90° **(d)** 45°

3. Light travels in: 1

(a) circles **(b)** curves always

(c) straight lines in a uniform medium **(d)** zig-zag always

4. The time for the same stars to rise at sunset again is the: 1

(a) tropical year **(b)** sidereal year **(c)** week **(d)** lunar month

5. The shape of the Moon visible from Earth changes through: 1

(a) eclipses only **(b)** seasons **(c)** tides only **(d)** phases

6. The phase when the whole lit face is visible is the: 1

(a) half moon **(b)** crescent **(c)** full moon **(d)** new moon

7. A habitat is a place that provides: 1

(a) the right conditions for an organism to live and grow

(b) only food

(c) only water

(d) only shelter from rain

8. The Sundarbans have the world's largest: 1

(a) deserts **(b)** glaciers **(c)** coral reefs **(d)** mangrove forests

9. Interconnected food chains form a: 1

(a) habitat **(b)** food web **(c)** biome **(d)** pyramid only

10. Transfer of pollen from anther to stigma is: 1

(a) pollination **(b)** budding **(c)** fertilisation **(d)** germination

11. Reproduction is essential for: 1

(a) making oxygen **(b)** continuity of life

(c) forming ozone **(d)** changing seasons

12. The hydrosphere consists of: 1

(a) air **(b)** water **(c)** rocks **(d)** living things

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 10–15 words each.

13. If the angle of incidence is 35°, what is the angle of reflection and the angle between incident and reflected rays? 2

14. Name the Moon's phases at the beginning and middle of the cycle. 2

15. What is a food web? 2

16. Why are decomposers called nature's cleaners? 2

17. What is the greenhouse effect? 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 30–40 words each.

18. Describe how a solar cooker makes use of a mirror. 3

19. Explain a luni-solar calendar with example. 3

20. What do you know about Vrikshayurveda and Kunapa Jala? 3

21. Explain why Earth's distance from the Sun is important. 3

22. Explain with an activity why a pencil appears bent in a glass of water. 3

OR

Why do vehicles use convex mirrors but dentists use concave mirrors? 3

23. How did people use the Moon's cycle to make lunar calendars? 3

OR

Describe the phases of the Moon during a lunar month with a sequence. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 80–100 words each.

24. A woman walks towards a large concave mirror. Describe what she sees and explain why. 5

OR

Compare plane, concave and convex mirrors in a table on shape, image, lateral inversion and use. 5

25. Explain mutualism, commensalism and parasitism with examples and a short note on each. 5

OR

Discuss the importance of the Sundarbans and the threats it faces. 5

END OF QUESTION PAPER Total: 50 Marks

MASTERMIND